

Roberto Echeverria Sosa

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EDUCATION

University of California San Diego (GPA: 3.68)

B.S. in Data Science | Machine Learning Track

La Jolla, CA

June 2027

EXPERIENCE

Software Engineering Intern

June 2026 - August 2026

Sumitomo Mitsui Banking Corporation (SMBC)

New York, NY

- Reduced analyst review time by ~90% (1+ hour to ~7 min per document) at ~\$0.05/doc with AI document intelligence infrastructure built on Databricks, since adopted as the firm's standard for all future document AI projects
- Delivered 95%+ accurate, confidence-scored field extraction for credit documents via a full-stack RAG system (React, FastAPI, Claude, Delta Lake), streamlining analyst review with structured Excel reports
- Resolved a document-size bottleneck by re-architecting the extraction pipeline with size-based routing, processing 143 lease agreements (27 fields each) in a 5-hour production run

Robotics Lab Fellow

May 2026 - Present

HDSI LAB 3.0, UC San Diego

La Jolla, CA

- Built the Robotic Flower, an interactive installation combining computer vision, TensorFlow, and robotics that blooms in real time when it detects a viewer smiling
- Trained a custom CNN for facial expression recognition, deployed on a Raspberry Pi to control an off-the-shelf robotic arm kit decorated as a flower, and programmed an Arduino LCD panel in C/C++ to display various messages

Autonomous Vehicle Engineer Intern

October 2024 - January 2025

Waymo Career Institute

Los Angeles, CA

- Designed a multi-sensor weatherproofing system for AVs in adverse weather, specifying hydrophobic/NIR sensor coatings, a heat dissipation circuit, and terrain-specific tires to maintain sensor accuracy across temperature extremes
- Proposed emergency sonar as a sensor to address low-visibility detection gaps in AVs and maintain safe navigation

LEADERSHIP

Consulting Director

January 2026 - Present

Data Science Student Society at UCSD

San Diego, CA

- Directed 6 teams of 4 engineers on quarterly consulting projects to deliver seamless UI/UX, robust codebases and automated deployments through GitHub Actions CI/CD pipelines for clients like the UCSD Center of Energy Research
- Established strategic partnerships between the club and companies like MongoDB, Playmerviv and IBM for consulting projects, pitching slide decks and presenting directly to management teams

Management Leadership for Tomorrow

Jan 2026 - Present

Career Preparation Fellow

Dallas, Texas

- Attended Deloitte University, networked with 40+ MLT partners and completed 20 hours of professional development

PROJECTS

Inference Scheduler for Self-Hosted LLM

- Built a request scheduler in front of a quantized LLM (Qwen3 8B) model on Jetson Orin Nano using FAST API, wrapping llama.cpp's API with an async queue and per-request timeouts to manage RAM constraints
- Exposed the service across devices via Tailscale with no public port exposure, sustaining 2 req/min at ~30s avg queue wait under concurrent load

NvironX | Hackathon Win

- Won 1st place out of 100 teams for Best Use of Impulse AI at DataHacks 2026 by delivering a model that produces risk scores across Community, Regulatory, Environmental, Utility, Governance, with mitigation recommendations
- Built a geospatial intelligence platform forecasting data center site viability from public records, utility, environmental, and social signals using Impulse AI for risk scoring and Claude for natural-language explanations

Credit Risk Predictor

- Cleaned 2M Lending Club loans (151 features), eliminating 60+ data leakage columns; built XGBoost credit default model achieving ROC AUC = 0.735 and 68% default recall on 20% base rate
- Deployed interactive Streamlit dashboard with real-time risk scoring, SHAP explainability, and dynamic waterfall visualizations

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, C, JavaScript, HTML, CSS, Git/GitHub, APIs

Software Tools: Pandas, Numpy, Matplotlib, Scikit-learn, Pytorch, Tensorflow, Apache Spark, AWS Cloud, Databricks

Domains: Cloud, ML/AI, GenAI, Data Structures & Algorithms, Full-Stack, Data Visualization, Agile Development

Hobbies: Arduino Robotics, LA Marathon (5 Hours), Chess (1K ELO), TikTok (105K followers), Guitar, Soccer